

Table A.9: Host prompt template files.

File	Purpose
base_notebook_assistant.txt	Core system persona and safety rules
jupyter_structure.txt	Cell index and markdown/code conventions
ask.txt	Ask mode: explanation only, no tools
code.txt	Code mode: Placement + python block format
agentic.txt	Agentic mode: two-phase query–implement
tool_calling/react_agent.txt	Tool-calling behaviour for Agentic rounds
tool_calling/tool_descriptions_autogen.txt	Auto-generated JSON schemas for tools

A.15 Sample Ask Response Characteristics

Although full model transcripts are omitted for brevity, reported Ask tests exhibited the following response structures:

- **Test 1.** Sectioned prose (loading → cleaning → features → modelling → evaluation) with variable names matching the Pakistan housing snapshot.
- **Test 2.** Numbered list addressing inputs, outputs, dependencies, purpose, and risks for cell 17.
- **Test 3.** Root-cause paragraph naming `NameError`, minimal fix (restore `df` from upstream), and impact statement—without a full replacement cell.

These structures align with the rubric keywords defined in `fyp_experiment_benchmarks_ask.json`.

A.16 Environment Setup

Minimum environment requirements for reproduction:

- Google Chrome with the project extension loaded (developer mode).
- Python 3.10+ with dependencies from the host requirements specification.
- Registered native messaging host (`register.ps1` on Windows).
- Valid `CEREBRAS_API_KEY` in `testing/host/.env`.